

Human Health Challenges Associated with a Manned Mars Mission

For many years, a manned mission to Mars has represented the next frontier in space exploration. To successfully land humans on the Martian surface, it will be necessary to overcome many significant challenges that require innovative solutions in the fields of engineering, physics, biology, chemistry, and psychology, among many others. Perhaps the most significant challenge that a manned mission to Mars will face is providing the adequate resources, techniques, and support that prioritize and protect the health and safety of the astronauts. After all, the success of a manned Mars mission is dependent on the crew being in sufficient health to land on and explore the Martian surface as well as return to Earth to provide insight and experience to allow humans to sustain an interplanetary presence. Many aspects of the mission are significantly dangerous to the health and well-being of the astronauts, and it is imperative that these risks are identified, controlled, and prevented to the extent required. How will a manned mission to Mars maintain human health and safety, and what challenges exist as an obstacle to this goal? In order to maintain crew health and safety, a vital challenge to mission success, it is necessary for a manned Mars expedition to develop a comprehensive healthcare system that includes fully autonomous medical diagnosis and treatment of ailments, the protection of the astronauts from the effects of the deep space atmosphere, and adequate methods of water management.

The prevalence of disease and injury will require a manned mission to Mars to develop an effective medical system capable of diagnosis and treatment of ailments in order to effectively protect human health. According to the ESA Study on the Survivability and Adaptation of Humans to Long-Duration Interplanetary and Planetary Environments (HUMEX study), the significant length of the Mars expedition increases the probability that the astronauts will encounter general diseases and injuries such as viruses, bacterial infections, open wounds, and fractures (Horneck and Comet 103). Additionally, both the HUMEX study and The Fourth Community Workshop on Achievability and Sustainability of Human Exploration of Mars (AM IV) make note of the detrimental physiological and psychological reactions in humans that result from sustained exposure to the deep space atmosphere (Horneck and Comet 103-104) (Charles et. al 74-85). Examples of these detrimental effects include increased risk of

bone fracture, carcinogenesis, cardiovascular disease, intracranial hypertension, psychiatric disorders, and renal stone formation (Charles et al. 74-85). Thus, even without consideration of the possibility of spacecraft failure, the crew of a manned Mars mission will have an elevated risk of developing medical ailments that will need to be addressed.

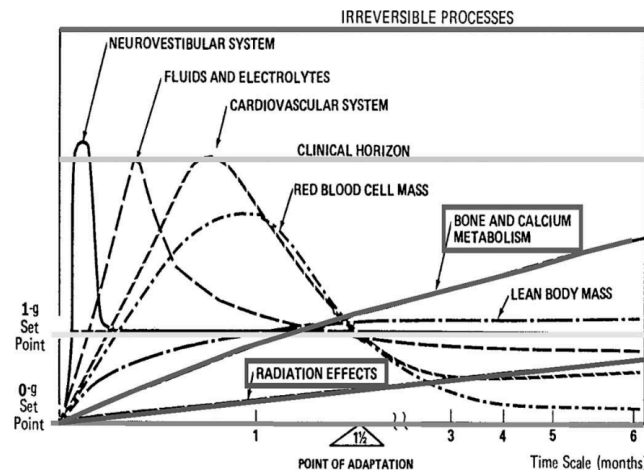


Figure 1: *Physiological Responses of Humans to the Deep Space Atmosphere (Horneck and Comet 106)*

The obstacles facing the development of an effective healthcare system capable of diagnosing and treating the plethora of ailments the astronauts may face include the barriers to communication and the extended mission duration. The HUMEX study estimates that telecommunication delays can reach 45 minutes (Horneck and Comet 106-107). In addition, AM IV asserts that the mission will be prone to communication blackouts due to orbital disruptions and solar particle interference (Charles et al. 78-79). Therefore, at certain times, the crew will be without live ground control, making telehealth practices and other medical support from Earth unfeasible for deep space exploration (Horneck and Comet 103). The significant duration of a manned mission to Mars and the distance that the spacecraft will be from Earth for the majority of the expedition will make fast mission abort and shipping medical supplies and equipment from Earth unattainable as well (Horneck and Comet 107). The inability of the crew to rely on medical support from Earth coupled with the likelihood that the astronauts will develop a variety of ailments leads both the HUMEX study and AM IV to agree that the mission will need to be able to develop a fully functional and autonomous medical diagnostic and therapeutic systems on board the

spacecraft, including surgical capacity (Horneck and Comet 103-107) (Charles et al. 78-79). Such a system would need to be able to analyze bone structure and impairments, identify pathogenic microorganisms, and track radiation dose rates to identify ailments, and use pulmonary life support systems as well as other therapeutic equipment to effectively monitor and protect the health of the astronauts (Horneck and Comet 103-107).

In developing an autonomous medical system on board the spacecraft, it is imperative for the mission to take into account size and mass constraints for medical supplies as well as the operational flexibility of equipment in fluctuating environmental conditions. An autonomous healthcare system aboard the spacecraft and in the Martian habitat will require sufficient medical equipment to allow for the effective diagnosis and treatment of the many ailments that may arise during the mission. However, according to AM IV, the volume of medical equipment and its mass are constrained due to limited space on the aircraft and in the Martian habitat, mass restrictions based on the predetermined quantity of fuel, and power limitations to ensure that other systems function effectively as well (Charles et al. 78-79). Therefore, AM IV argues that it is vital to ascertain the most essential and practical medical equipment, determined by a variety of factors including likelihood of use and functionality in the deep space environment (Charles et al. 78-79). The HUMEX study also asserts that further technological development of healthcare equipment used on Earth is required to assure functionality in fluctuating gravitational conditions and in the presence of higher carbon dioxide levels caused by the deep space environment (Horneck and Comet 106-107). For example, equipment such as pulmonary life support systems, used for decompression sickness, anesthesia, and in surgery, will need to be adapted for use in microgravity levels during the transit phase of the mission and high gravity levels upon aerocapture which can reach up to 6g (Horneck and Comet 106-107). Since developing an autonomous healthcare system in the spacecraft and in the Martian habitat is necessary for a manned mission to Mars, it will be vital to address the logistical challenges and further technological innovation needed in order to provide equipment capable of medical diagnosis and treatment.

Several aspects of the deep space environment will require the comprehensive healthcare system to incorporate additional layers of protection for the astronauts on a Mars mission. Once the crew exit Earth's magnetosphere, they will face a prolonged exposure to higher levels of radiation according to S.T. Suess, a physicist at the Marshall Space Flight Center in Huntsville, Alabama (Suess 556). There are two types of space radiation that are of most concern to a manned Mars mission. The first type of radiation that the astronauts will face in the deep space environment are caused in part by solar particle events (SPE) which include flares and filaments which can last days (Suess 556-557). Another type of space radiation is galactic cosmic rays (GCR) which, according to Slaba et al. in "Radiation Shield Optimization on Mars", are a higher concern than SPE due to the current inability to effectively protect against that type of radiation (Slaba et al. 1). Additionally, in their paper "A Hypothesis on Biological Protection from Space Radiation Through the Use of New Therapeutic Gasses", Schoenfeld et al. assert that both types of space radiation have a high linear energy transfer rate, increasing DNA double strand breaks which are difficult for natural repair mechanisms to correct (Schoenfeld et al. 1-2). Suess and AM IV list the detrimental effects of radiation exposure which include disruption of radio communication as well as cardiovascular disease, degenerative tissue effects, and carcinogenesis which can lead to death (Suess 556) (Charles et al. 76-78). Since prolonged radiation exposure will significantly increase the likelihood that the astronauts will develop ailments on a manned Mars mission, it is necessary to develop methods and technologies that monitor, mitigate, and prevent these effects to maintain the health of the crew.

To accurately protect against radiation exposure, it is important to incorporate systems that monitor and provide accurate information about the environment in which the crew is operating. Monitoring radiation dose rates will take several forms. First, Suess argues that the mission will need to monitor solar changes to accurately predict flares and filaments, two types of SPEs (Suess 556-560). The flux of particles from SPEs will take between fifteen and thirty minutes to reach the astronauts, so accurate identification of these events provides a window in which the astronauts are able to react in ways to lower radiation exposure (Suess 556-557). Second, both the HUMEX study and AM IV emphasize the importance of developing radiation dose rate monitoring technology that identify markers and allow for a

comprehensive assessment of risk (Horneck and Comet 103-107) (Charles et al. 76-79). This multi-faceted approach to radiation monitoring will enable the astronauts to determine radiation levels and risk in a timely manner which will allow them to make informed decisions about the mission in order to protect their health and safety and effectively diagnose ailments that result from exposure.

In conjunction with monitoring the environment in order to reduce the occurrence of radiation-related ailments, it is necessary for a manned mission to Mars to incorporate effective shielding methods to protect against exposure. For SPEs, both the HUMEX study and Slaba et al. posit the need for radiation shields to protect the astronauts (Horneck and Comet 103) (Slaba et al. 1-6). Optimizing the materials used for shielding during SPEs has shown to be effective at reducing radiation dose rates (Slaba et al. 1-6). If accurate methods of SPE prediction as described in the previous paragraph are deployed on a deep space mission, it will allow the crew to effectively utilize shielding methods, preventing negative outcomes associated with prolonged radiation exposure.

Preventing and mitigating the effects caused by exposure to GCRs provides a much greater challenge for a mission to Mars. According to Slaba et al., shielding methods have been shown to be incapable of preventing radiation exposure due to GCRs, and in certain cases enhance dose rates due to secondary radiation (Slaba et al. 1-6). Secondary radiation is caused when GCRs undergo nuclear fragmentation during interaction with the shields (Schoenfeld et al. 1-2). The result of this process is the development of energetic neutrons which emit radiation (Slaba et al. 1-2). In a study conducted by Slaba et al., it was shown that polyethylene is capable of marginally diminishing secondary radiation by preventing the production of energetic neutrons more effectively than aluminum, making it more desirable (Slaba et al. 1-6). However, as previously stated, shielding methods are not effective enough to significantly reduce the incidence of radiation illnesses even with the use of polyethylene (Slaba et al. 1-6). Another method being considered for the prevention of radiation exposure is through in-situ resource utilization (ISRU), a process of using resources found in extraterrestrial environments to develop necessary materials. In their paper “Opportunities and Strategies for Testing and Infusing ISRU in the Evolvable Mars Campaign”, Mueller et al. advocate for utilizing asteroidal material to develop radiation

shielding panels for the spacecraft and regolith on the Martian surface to protect ground habitats (Mueller et al. 3-5). However, Slaba et al. argues that ISRU is not a practical means to prevent radiation exposure because it will require large scale excavation of Martian regolith due to the significant thickness requirements (Slaba et al. 6). A third option for enhanced protection from radiation to deal with the effects of GCRs is using biological methods to mitigate the negative effects of radiation exposure. According to Schoenfeld et al., therapeutic gasses can be used to suppress the decomposition of water in cells that results from radiation exposure, a process called radiolysis (Schoenfeld et al. 2-6). Antioxidants within the body respond to radiolysis by inducing the reformation of water (Schoenfeld et al. 2-6). High levels of radiation such as the astronauts will face on a manned Mars mission will cause an imbalance between the levels of antioxidants and byproducts of water decomposition (Schoenfeld et al. 2-6). Once this imbalance occurs, the body's cells undergo oxidative stress resulting in the symptoms associated with radiation exposure (Schoenfeld et al. 2-6). To mitigate these effects, Schoenfeld et al. posit the use of the therapeutic gas hydrogen which will stimulate the reformation process of water within the cells and suppress the effects of the byproducts of radiolysis, preventing the aforementioned imbalance and oxidative stress caused by radiation (Schoenfeld et al. 2-6). Furthermore, Schoenfeld et al. argue that genetic modification of certain molecular targets such as the Apolipoprotein E genotype reduce the negative effects of radiation, such as preventing fibrosis in tissue (Schoenfeld et al. 5-6). A manned expedition to Mars will need to discern the most practical and effective system for radiation shielding to prevent related illnesses in the astronauts in order to maintain human health and prevent performance decreases to enhance the odds of crew survival and mission success.

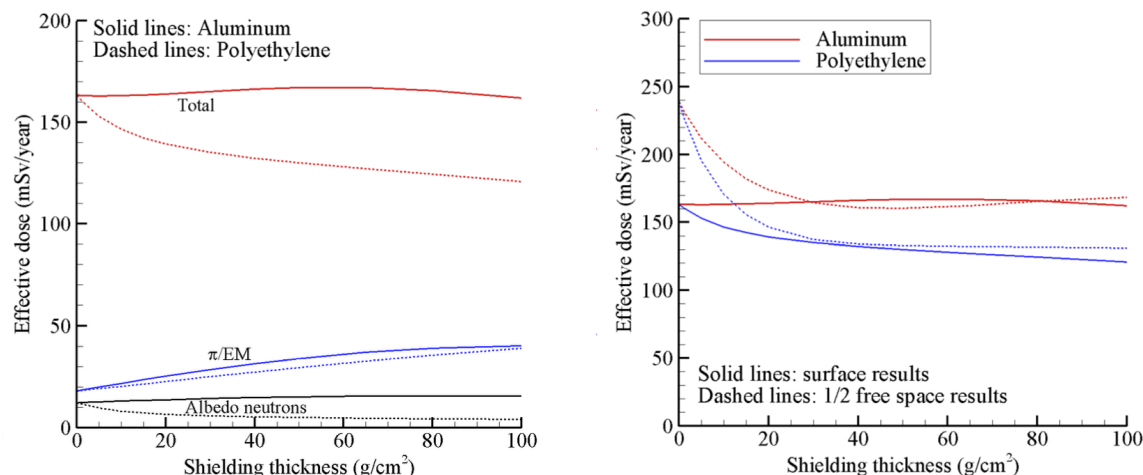


Figure 2: Effectiveness of Radiation Shields Composed of Polyethylene and Aluminum (Slaba et al. 5-6)

Another aspect of the deep space environment that a comprehensive healthcare system will need to address is the effects of prolonged levels of microgravity that the astronauts will experience. According to the scholarly article “Artificial Gravity in Mars Orbit for Crew Acclimation” by the engineer Justin Rowe, the HUMEX study, and AM IV, long term stays in microgravity, such as a deep space expedition to Mars, causes bone demineralization (Rowe 1-6) (Horneck and Comet 103) (Charles et al. 79-81). On Earth, bones are strengthened by minerals, such as calcium. This process of bone mineralization is the body’s response to the weight forces exerted on the bones during motor movements. Microgravity environments decondition the bone mineralization process because the forces acting upon the skeletal system are significantly reduced, so bones require less calcium and other minerals for strength. The result is bone demineralization which can cause debilitating health effects such as osteoporosis and bone fractures when the astronauts are exposed to higher gravitational levels during aerocapture and landing as well as on the Martian surface (Horneck and Comet 103) (Charles et al. 79-81). According to the HUMEX study, rates of up to 50% calcium loss in the region around the pelvis are predicted for the astronauts which far exceeds the threshold of 15% loss that is considered to put a human at significant risk for bone fracture (Horneck and Comet 103). Rowe states that in addition to deterioration of bone composition, long term stays in microgravity can cause declining vestibular sensorimotor capabilities and decrease performance of simple tasks (Rowe 6-7). The health risks to the astronauts of bone

demineralization caused by the microgravity pose a significant challenge because the debilitating nature of these ailments can reduce the functioning of the crew and require larger amounts of time be spent on medical treatment of these ailments which put the survival odds of the astronauts and mission at risk. Thus, it is vital for the manned Mars mission to establish methods to prevent bone demineralization and its negative effects.

In order to account for the negative effects of long term exposure to microgravity, it will be important to incorporate artificial gravity into the comprehensive healthcare system. The HUMEX study and AM IV both express the importance of utilizing equipment to observe bone composition, mineral levels, and other important biomarkers (Horneck and Comet 106-107) (Charles et al. 79-81). However, as AM IV notes, bone mineralization recovers to acceptable levels before bone strength (Charles et al. 80). Therefore, it is necessary for a manned Mars mission to prevent the bone demineralization process which can be accomplished by the implementation of artificial gravity. Rowe asserts that the most effective artificial gravity system is through rotation of the spacecraft, a view corroborated by the HUMEX study (Rowe 1-4) (Horneck and Comet 103-104). The centrifugal acceleration vector is the artificial acceleration due to gravity in this scenario (Rowe 1). The development of a rotational system of artificial gravity must take into account several constraints that result from the nature of a manned mission to Mars. Crew comfort must be considered to provide the astronauts with an environment in which they are able to effectively perform required tasks vital to mission success. According to Rowe, an obstacle facing the implementation of a rotating artificial gravity system is the Coriolis effect (Rowe 4-5). On the spacecraft, the Coriolis effect will occur with the movement of the crew inside the rotating reference frame of the spacecraft, increasing the difficulty of locomotion and translation and causing side effects such as nausea (Rowe 4-5). Thus, a rotating artificial gravitational system must seek to reduce the discomfort to the crew and impacts on performance of tasks that results from the Coriolis effect which is often achieved by implementing slower rates of rotation. The ideal rate of rotation that would not alter the astronauts' locomotion or translation would be 2 RPM, but such a system would require a prohibitively large radius in order to maintain a desirable centrifugal acceleration (Rowe 4-5). However, a short radius with a

relatively large rotation rate will increase the aforementioned discomfort among the astronauts that results from the Coriolis effect (Rowe 4-5). Thus, Rowe argues for the implementation of a moderate rotation rate and radius that will require the astronauts to adapt, but will provide for crew comfort to the extent possible within the constraints of the mission (Rowe 4-5). Also, the floors of the spacecraft will need to be perpendicular to the radius of rotation in order for the centrifugal force to act as artificial gravity (Rowe 5-7). This leads Rowe to believe that a vertical orientation for the spacecraft is ideal because it ensures that the floors remain perpendicular to the radius of rotation (Rowe 5-7). Thus, developing a rotating artificial gravity system is a significant design and engineering challenge that will need to be addressed prior to a manned Mars mission in order to prevent the negative effects of microgravity that the astronauts will face in the deep space environment.

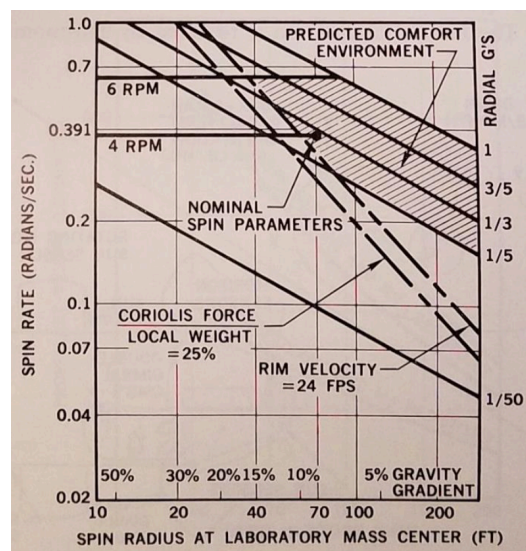


Figure 3: Constraints on a Rotating Artificial Gravity System to Provide for Crew Comfort (Rowe 5)

The final aspect that a comprehensive healthcare system on a manned mission to Mars will need to incorporate is effective water management. Water is an essential resource required by the astronauts for survival, and it can be used in other tasks crucial to the health of the crew including cleaning, as flush water, and growing or rehydrating food if necessary (Jones et al. 1-4). Transporting these resources from Earth poses many challenges to the mission including the risk of contamination, long periods of storage, and volume and mass constraints that limit the amount able to be launched from Earth (Jones et al. 1-4).

Additionally, the significant distance between Earth and the astronauts for a significant portion of the mission will make launching these resources unfeasible (Horneck and Comet 106-107). Thus, it will be necessary to develop systems of resource management that address these challenges because an adequate supply and quality of water is crucial to ensure the survival of the crew and increase the odds of mission success.

Water acquisition methods, including recycling systems and ISRU, are able to reduce the launch mass of resources and have the potential to provide a sufficient supply to the astronauts. In the scholarly article “Developing the Water Supply System for Travel to Mars”, Jones et al. assert that the three sources for recycling water include from condensate, leftover liquid from cleaning, and urine and toilet water (Jones et al. 1-4). Steele et al. corroborate this assertion in their paper “Antimicrobials for Water Systems in Manned Spaceflight - Past, Present, and Future Applications and Challenges” (Steele et al. 1-3). According to Jones et al., condensate is recycled by removing humidity from the air via a condensing heat exchanger (Jones et al. 14-15). The ISS Water Processor Assembly, a multifiltration condensate recycling system, could be considered for use on a Mars mission if further development can reduce the size and mass and increase its reliability. (Jones et al. 15). One method to increase the functionality of the ISS WPA is to incorporate parallel water processing systems which despite the higher upfront costs, their decentralization results in increased reliability and decreased likelihood of total system failure due to the relative independence of each subsystem (Jones et al. 14). In order to recycle water used for cleaning and flushing as well as urine, Jones et al. maintain that it is necessary to utilize a phase change technology such as evaporation or distillation (Jones et al. 15). According to Jones et al., sufficient amounts of water are available in the forms of condensate, urine, and waste water to provide the crew an adequate supply for hydration and further cleaning (Jones et al. 4)

Mars transit water system requirements.

Requirements	Value
General	
Number of crew	4
Mission duration	400-500 days
Quiescent interval	400-550 days
Pr(LOC)	< 0.001
Water consumption	
Drinking and food preparation water	2.38 kg/CM-d
Urine flush water	0.50 kg/CM-d
Wash water	1.29 kg/CM-d
Total water	4.17 kg/CM-d
Waste water	
Respiration and perspiration condensate	2.28 kg/CM-d
Urine and flush water	2.00 kg/CM-d
Used wash water	1.29 kg/CM-d
Total waste water	5.57 kg/CM-d

CM - crew member, d - day

Figure 4: Water System Requirements for a manned Mars mission (Jones et al. 4)

Another method to reduce the mass of water and other resources launched from Earth is ISRU. According to Mueller et al., water can be acquired through ISRU by drilling into polar ice caps or the process of desorption by heating hydrated minerals found in regolith on the Martian surface (Mueller et al. 6). The prevalence of hydrated minerals on the Martian surface makes ISRU an appealing option, however, as Mueller et al. notes, development and testing of technologies capable of converting materials found during the expedition into water are a significant barrier to the effective implementation of an ISRU water supply system.

In order to use water obtained through recycling or ISRU, it is necessary to develop quality control mechanisms that will prevent microbiological growth and other contaminants from manifesting in the water that could be detrimental to astronaut health and hinder the odds of mission success. The scholarly article “Antimicrobials for Water Systems in Manned Spaceflight - Past, Present, and Future Applications and Challenges” develops criteria regarding the selection of an antimicrobial that could be incorporated into a water management system to prevent microbiological growth (Steele et al. 2-3). This criteria includes human health, toxicity, flammability, effectiveness, methods to add additional antimicrobial, long periods of system disuse, and microbial resistance and mutation over time (Steele et al. 2-3). After their analysis of antimicrobials used in previous space expeditions, Steele et al. argue that silver is the ideal candidate (Steele et al. 13). Silver has many desirable characteristics for antimicrobial use in a water management system including its harmlessness to human health at the levels required to

suppress microbial growth caused in part by the fact that the body eliminates 99% of the substance (Steele et al. 13). However, the water management system will need to account for several factors that impede silver's antimicrobial capability including temperature, pH, and colloidal particles (Steele et al. 13). Therefore, it will be vital for the mission to control these parameters in order to increase silver's effectiveness. Additionally, Steele et al. notes that silver readily reacts with materials which will require a dose of at least 50 parts per billion be added to the water (Steele et al. 13). This dose rate will allow water concentrations to remain at a high enough level regardless of water's reactivity with other materials in order to prevent pathogenic bacterial growth and microbial mutation and resistance (Steele et al. 13). Another challenge in utilizing silver is the effective duration antimicrobial due to long periods of dormancy that result from the extended mission duration (Steele et al. 13). According to Steele et al., it has been shown that the ion complex silver dihydrogen citrate (SDC) has the capability to decontaminate water for up to five years which results from its ability to denature DNA in the cytoplasm of microbials (Steele et al. 13). In order to implement an effective water management system, a manned mission to Mars will need to utilize antimicrobials such as silver to effectively suppress microbial growth and other contamination in order to ensure the health, safety, and capability of the astronauts are maintained.

To develop a comprehensive healthcare system on a manned Mars mission, it is imperative to be able to fully and autonomously diagnose and treat medical ailments, protect the astronauts from the effects of the deep space atmosphere, and adequately manage the water supply. Addressing these topics is vital to maintain the health and safety of the astronauts because the systems that are incorporated will address the significant health concerns that arise from deep space expeditions. In addition to the moral duty of the mission to prioritize the preservation of life, maintaining human health standards is necessary for mission success and any resulting scientific advancement including data collection and determining the future efficacy of Mars expeditions. With the central focus of a manned Mars mission being human exploration, maintaining the health and safety of the crew is the most important aspect in order to achieve that vision.

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